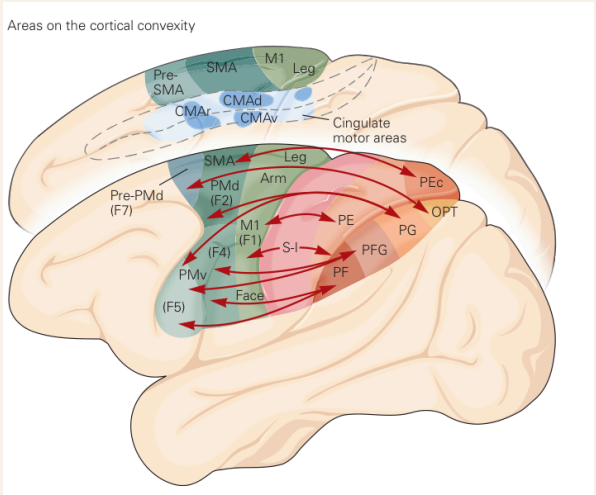
PHYSIOLOGY OF THE MOTOR SYSTEM

From an integrated body–environment view to action

Lecture 02

A connected cortical network



Kandel 6e, Fig. 34–2B, p. 820 · macaque cortical convexity

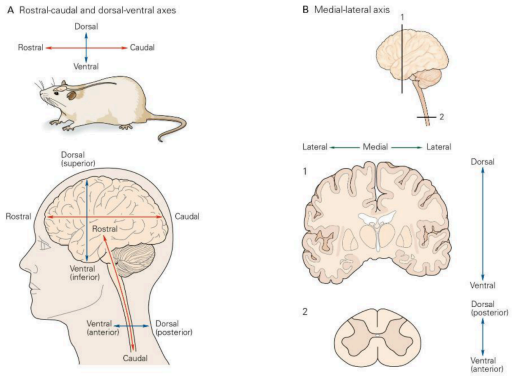
> The cortical motor network includes parietal, premotor and primary motor regions.

LECTURE NOTE

PPC, premotor cortex and M1 occupy connected territories, with reciprocal routes between them. This map provides the anatomical setting for the experiments; its arrows do not require every action to pass through a fixed serial relay.

YOUR NOTES

Read the anatomical directions



Prior L01 direction slide · Kandel Fig. 1–2 · archived 2025–2026 deck, p. 26

- Rostral: towards the nose / Caudal: towards the tail
- Medial: towards the midline / Lateral: away from the midline
- Dorsal: towards the back / Ventral: towards the belly

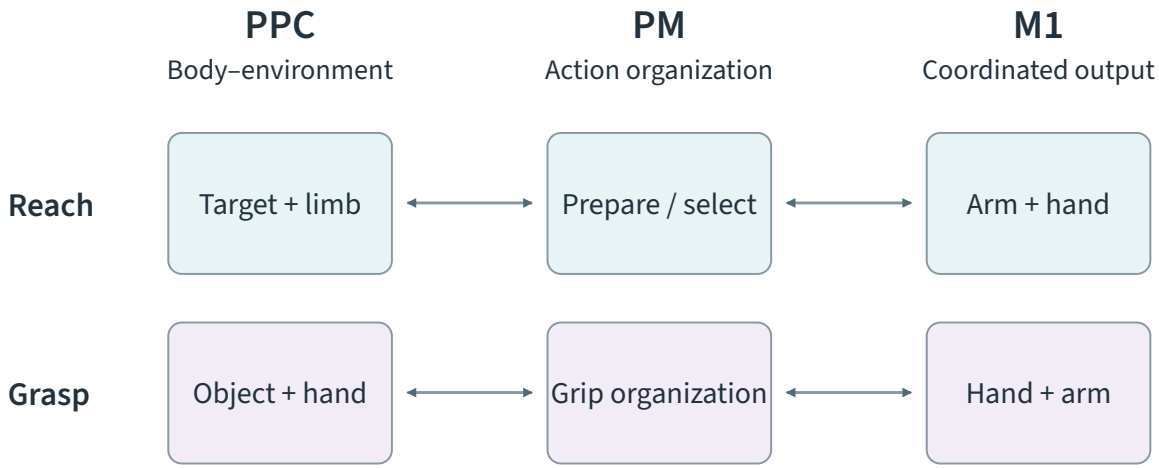
A.02 2 / 50

LECTURE NOTE

In the cerebrum, rostral and caudal point toward the frontal and occipital poles; medial and lateral refer to distance from the midline. Proximal and distal instead describe positions along the limb, so an effector description must not be confused with a cortical direction.

YOUR NOTES

From body–environment relationships to action



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

- Physiological distinctions between processing stage and movement type.
- Interacting contributions, not a compulsory relay.

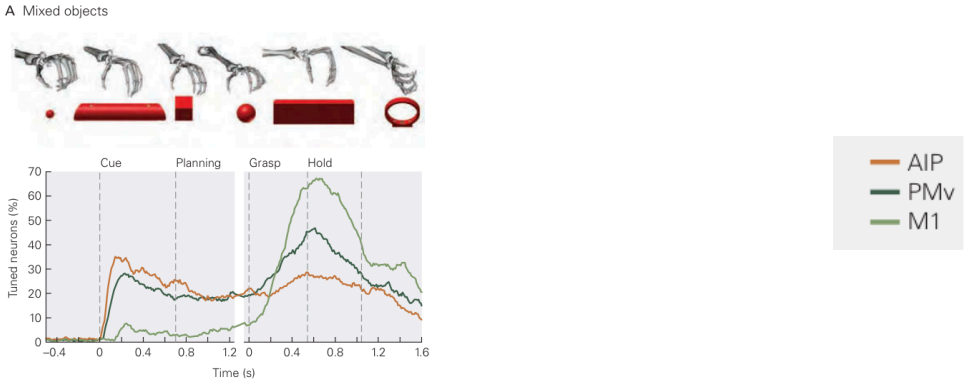
A.03 3 / 50

LECTURE NOTE

The lecture follows complementary contributions within a connected controller: relating body and environment, organizing an action, and coordinating output. Reach and grasp recur as examples so that the regional explanations can be connected rather than learned as separate inventories.

YOUR NOTES

Which information differs across the grasping circuit?



Kandel 6e, Fig. 34–15A, p. 840 · mixed objects; original shared legend

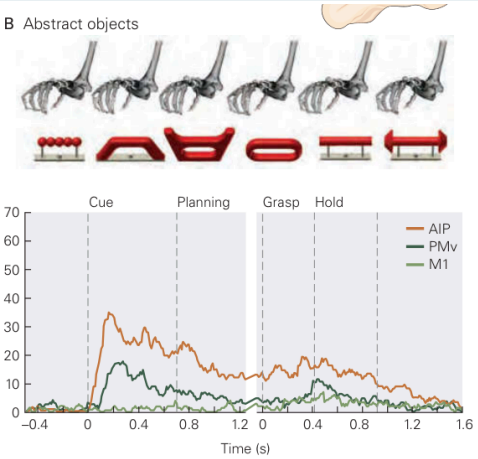
- Cue: Object appears
- Planning: Prepare the grasp
- Grasp: Reach, then grasp the object
- Hold: Maintain the grasp

LECTURE NOTE

Mixed objects vary in both visual shape and the hand configuration required to grasp them. AIP selectivity is prominent during viewing and planning, while PMv and M1 differentiate the actions during grasp and hold; this comparison alone cannot separate shape from grasp information.

YOUR NOTES

Different objects can require similar grasps



Kandel 6e, Fig. 34–15B, p. 840 · abstract objects

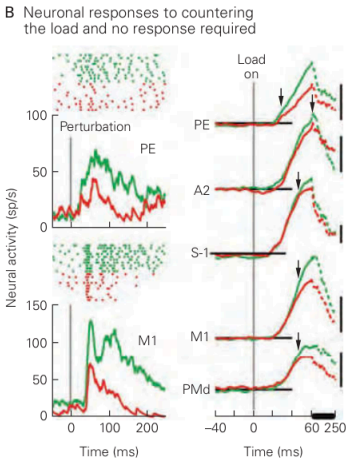
- Different shapes, similar grips

LECTURE NOTE

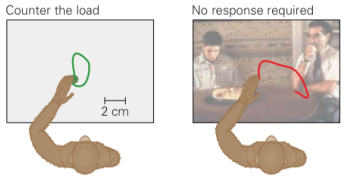
Keeping the grasp similar while changing appearance separates visual shape sensitivity from differences associated with grasping. AIP retains object selectivity, whereas PMv and M1 differentiate these conditions much less; together with the mixed-object result, this supports different regional emphases within a connected network, without proving a fixed serial relay.

YOUR NOTES

Task engagement changes feedback responses



Kandel 6e, Fig. 34–23A/B, p. 853 · counter the load versus no response



- Unexpected arm load
- Green: correction to center
- Red: movie; no correction required

- Feedback responses across regions
- Earliest task modulation: PE
- Arrows: first condition difference

PE: superior parietal area 5
A2: somatosensory area 2
S1: primary somatosensory areas 1 and 3

M1: primary motor cortex
PMd: dorsal premotor cortex

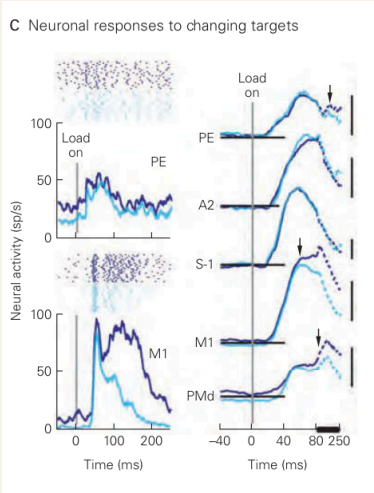
A.06 6 / 50

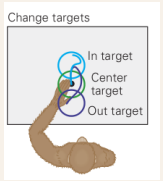
LECTURE NOTE

The robot applies an unexpected arm load while the monkey either corrects to the central target or watches a movie without needing to correct. Feedback evokes activity across the recorded network, but task modulation appears earliest in parietal PE; the arrows mark condition differences, not initial responses. This extends the grasp comparison to task-dependent feedback within the connected controller.

YOUR NOTES

Corrective demands change the regional timing





- Arm load: cue to a new target
- Cyan: push toward target (IN)
- Purple: push away (OUT)

- Earlier target modulation in M1
- Later parietal target differences
- Timing dependent on the comparison

Kandel 6e, Fig. 34–23A/C, p. 853 · disturbance toward versus away from the new target

PE: superior parietal area 5
A2: somatosensory area 2
S1: primary somatosensory areas 1 and 3

M1: primary motor cortex
PMd: dorsal premotor cortex

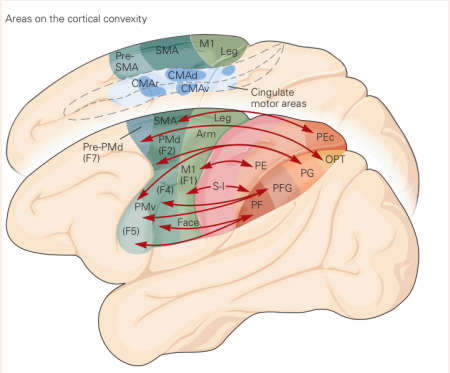
A.07 7 / 50

LECTURE NOTE

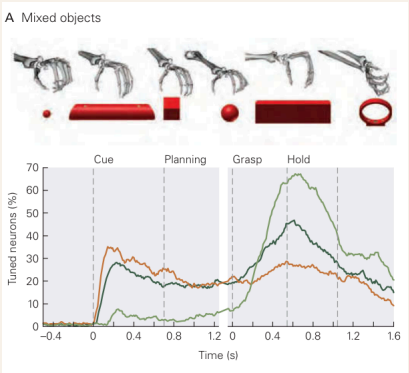
When the load indicates a new target, M1 distinguishes the corrective demands earlier than the parietal regions in this comparison. Together with the preceding task-engagement contrast, this shows why regional contributions cannot be ordered by a single universal feedback latency.

YOUR NOTES

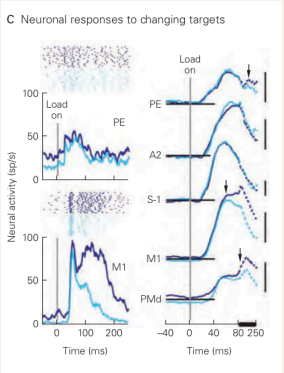
Section A summary: one network, shifting emphases



Connected network



Shifting emphasis



Task-dependent feedback

TAKE HOME One connected network; regional emphasis depends on the task.

Kandel 6e, Figs. 34–2B, p. 820; 34–15A, p. 840; 34–23C, p. 853

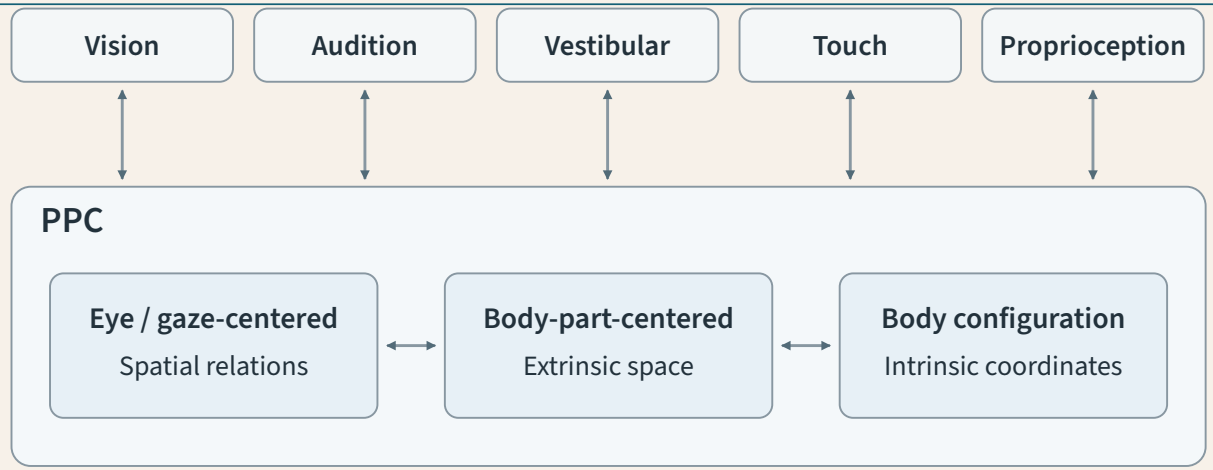
A.08 8 / 50

LECTURE NOTE

Section A located parietal, premotor and primary motor cortex as one connected network and showed that its regions differ in emphasis rather than forming a fixed relay: object and grasp information shift across AIP, PMv and M1, and feedback timing depends on the task. The next sections examine each region's contribution, beginning with PPC.

YOUR NOTES

B · Integrating body and environment



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

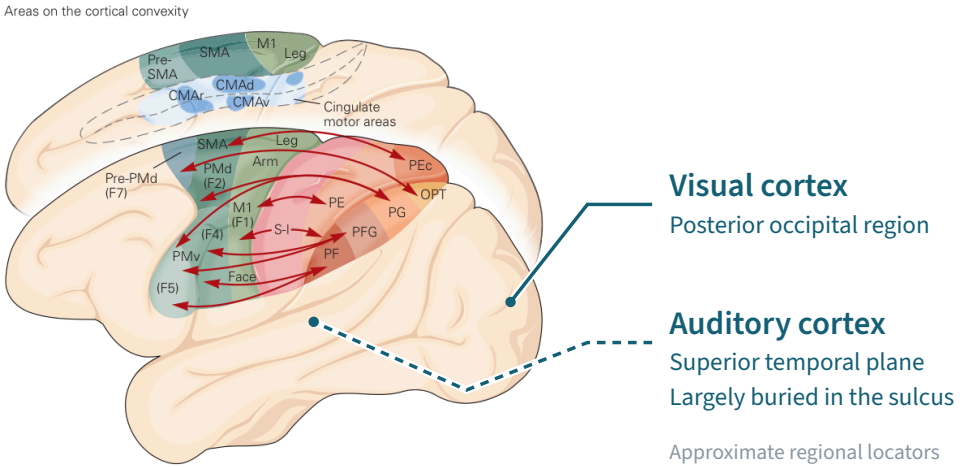
- Parietal organization reflects the organization of its sensory inputs.
- Linked, mixed representations.

LECTURE NOTE

PPC combines several sensory sources with body and motor-related information. The three internal networks summarize complementary descriptions of body–environment relationships; they are teaching abstractions rather than three exhaustive, anatomically segregated processing stages.

YOUR NOTES

Locate the sensory sources around PPC



Kandel 6e, Fig. 34–2B, p. 820 · added approximate visual/auditory locators; auditory region largely buried.

> Posterior parietal cortex integrates different sensory signals to identify possible targets of action.

B.02 10 / 50

LECTURE NOTE

Return to PPC on the anatomical map while locating visual and auditory sources alongside somatic information. Broad regional locators orient the explanation; the anatomical drawing by itself does not establish multisensory integration or assign each coordinate description to one named area.

YOUR NOTES

Same reach, different gaze



$$(T - G) - (H - G) = T - H$$

G: gaze · H: hand · T: target

Kandel 6e, Fig. 34–5C, conditions 1 and 3, pp. 826–827 · task drawings only; geometry illustrative

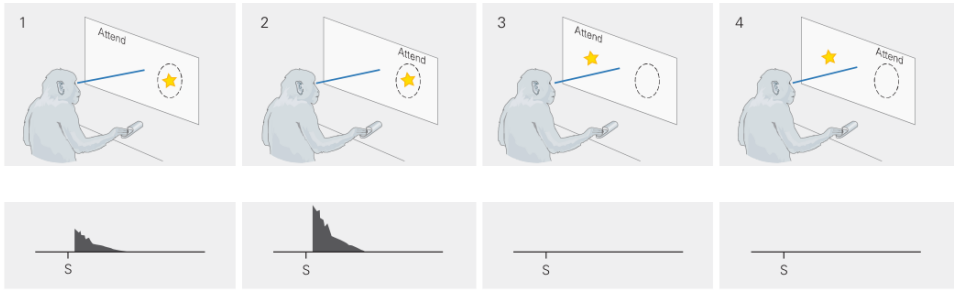
B.03 11 / 50

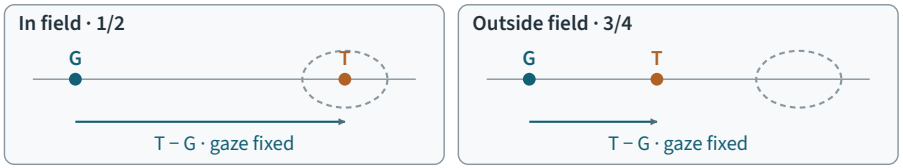
LECTURE NOTE

Conditions 1 and 3 show the same physical reach while gaze shifts from the target to another location. The schematic relates that behavior to gaze-, hand- and target-relative descriptions: the gaze-relative quantities change, but their difference preserves the hand–target relationship. Later neural comparisons ask which of these relationships activity follows; this introductory view shows no neural responses.

YOUR NOTES

LIP · visual location and attention

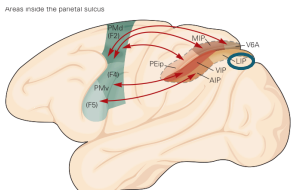




G: gaze · T: flashed stimulus · dotted oval: receptive field

Kandel 6e, Figs. 34–5A, pp. 826–827; 34–2B, p. 820 · schematic illustrative

Lateral intraparietal area



- Dotted circle: field
- 1 ↔ 2: attention
- 1/2 ↔ 3/4: location

- In field: response
- Attended: stronger
- Outside: little response

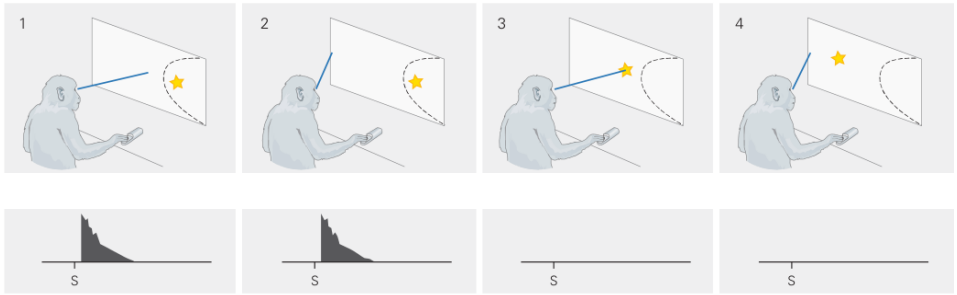
B.04 12 / 50

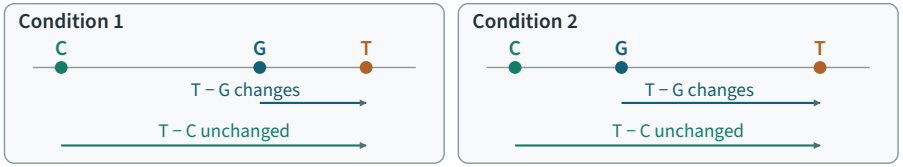
LECTURE NOTE

LIP relates a visual stimulus to a gaze-relative receptive field, with stronger firing when attention is directed to an in-field stimulus. Compare conditions 1 and 2 for attention and 1/2 with 3/4 for field location; gaze itself stays fixed in these drawings, so this example adds visual selectivity to the spatial relationships introduced earlier.

YOUR NOTES

VIP · location relative to the head

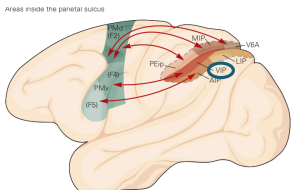




C: head · G: gaze · T: flashed stimulus · geometry of 1/2

Kandel 6e, Figs. 34–5B, pp. 826–827; 34–2B, p. 820 · schematic illustrative

Ventral intraparietal area



- 1 ↔ 2: same head-relative location
- 1 ↔ 4: similar retinal offset

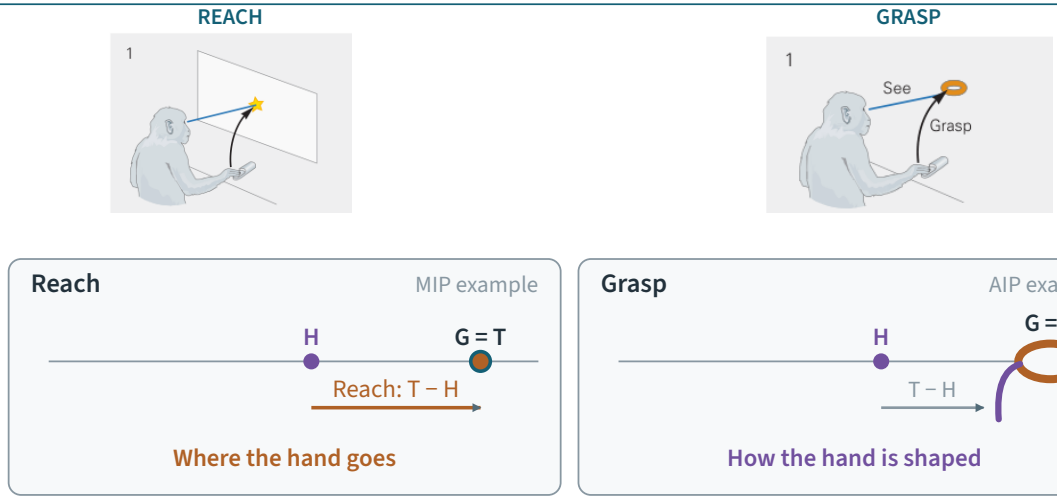
- 1/2: similar response
- 1/4: different response
- Head-relative sensitivity

LECTURE NOTE

This VIP neuron responds to a visual stimulus at a particular location relative to the head across changes in gaze. Conditions 1 and 2 preserve that head-relative location, whereas 1 and 4 have similar retinal offsets but different responses; the comparison extends the gaze-relative description to another body reference without proving one universal parietal coordinate frame.

YOUR NOTES

Reach and grasp: the hand acts on the target



G: gaze · H: hand · T: target or object · illustrative geometry

Acting needs the target relative to the hand, and the hand shaped to the object.

Kandel 6e, Fig. 34–5C and D, condition 1, pp. 826–827 · task drawings only; geometry illustrative

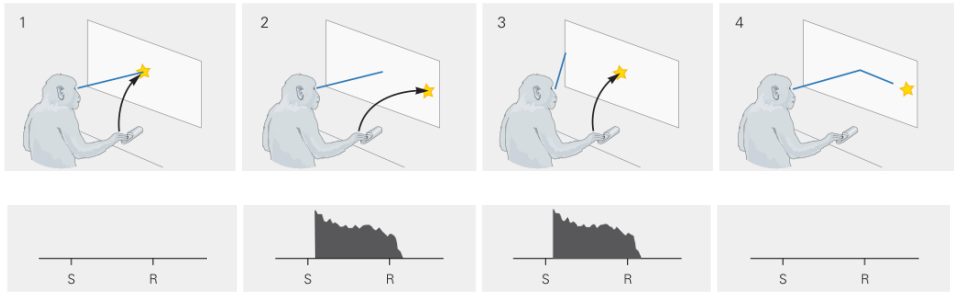
B.06 14 / 50

LECTURE NOTE

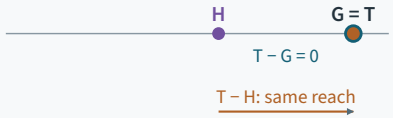
LIP and VIP located visual stimuli relative to the eye or head; in the next two experiments the monkey acts with its arm and hand. Reaching needs the target's location relative to the hand, whereas grasping also needs the hand shaped to the object, so MIP and AIP ask which of these relationships parietal activity follows.

YOUR NOTES

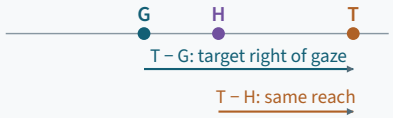
MIP · reach direction relative to gaze



1 · Gaze at target



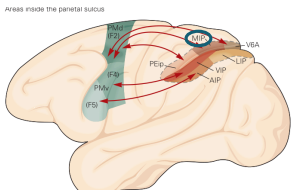
3 · Gaze elsewhere



G: gaze · H: hand · T: reach target · geometry of 1/3

Kandel 6e, Figs. 34–5C, pp. 826–827; 34–2B, p. 820 · schematic illustrative

Medial intraparietal area



- 1 ↔ 3: same reach, different gaze
- 2 ↔ 3: different reach
- 4: eye movement, no reach

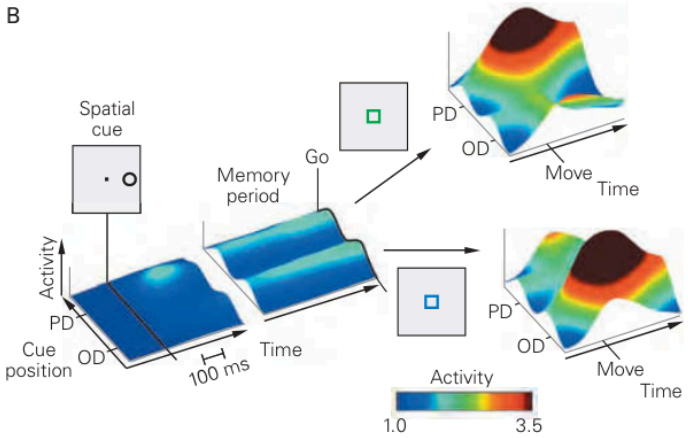
- Only 2 and 3 respond
- Both: target right of gaze
- No reach: no response

LECTURE NOTE

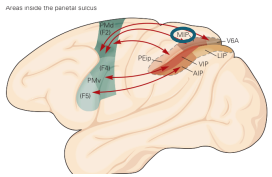
This MIP neuron fires when the reach target lies right of gaze (2, 3), but not for the same physical reach to a fixated target (1) or an eye movement alone (4). It answers the earlier coordinate comparison: reach-related activity can follow the target's direction relative to gaze rather than the arm's physical path.

YOUR NOTES

PRR · potential reaches in parietal cortex



Parietal reach region (≈ MIP)



- One spatial cue
- Later color cue: reach to it (PD) or opposite (OD)

- Delay: both directions present
- Then the instructed reach
- Options held before choice

Kandel 6e, Fig. 34–8B, p. 832; PRR likely corresponds to MIP, p. 825; 34–2B, p. 820 · remembered target and later reach instruction

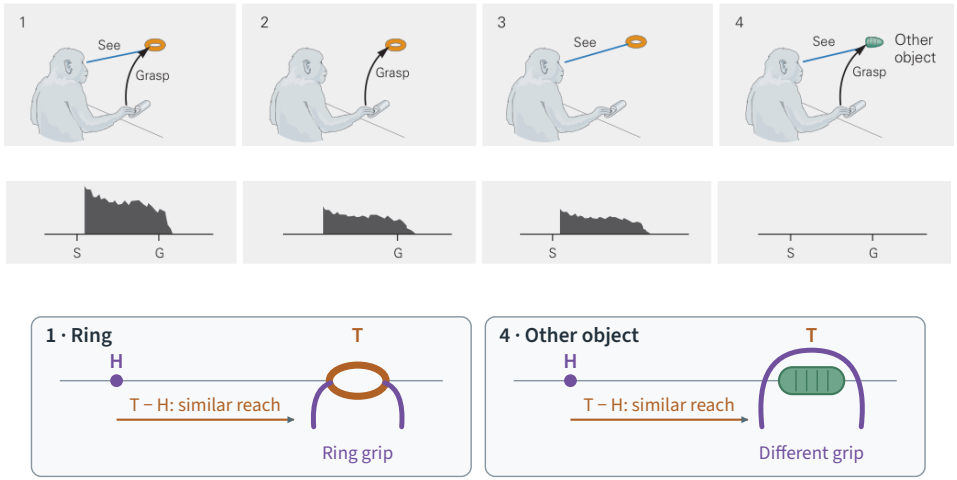
B.08 16 / 50

LECTURE NOTE

In the parietal reach region, which likely corresponds to MIP, one remembered spatial cue is followed by an instruction to reach toward or away from it. Population activity holds both potential reach directions until the instruction, so parietal reach activity concerns possible actions as well as target location; premotor cortex will show a related result.

YOUR NOTES

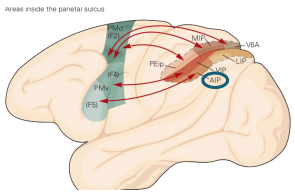
AIP · object shape for grasping



H: hand · T: object location · similar position, different shape and grip

Kandel 6e, Figs. 34–5D, pp. 826–827; 34–2B, p. 820 · schematic illustrative

Anterior intraparietal area



- 1 ↔ 4: similar reach, other object
- 2: grasp in darkness
- 3: view, no grasp

- Other object: none
- Dark grasp: response
- View only: response

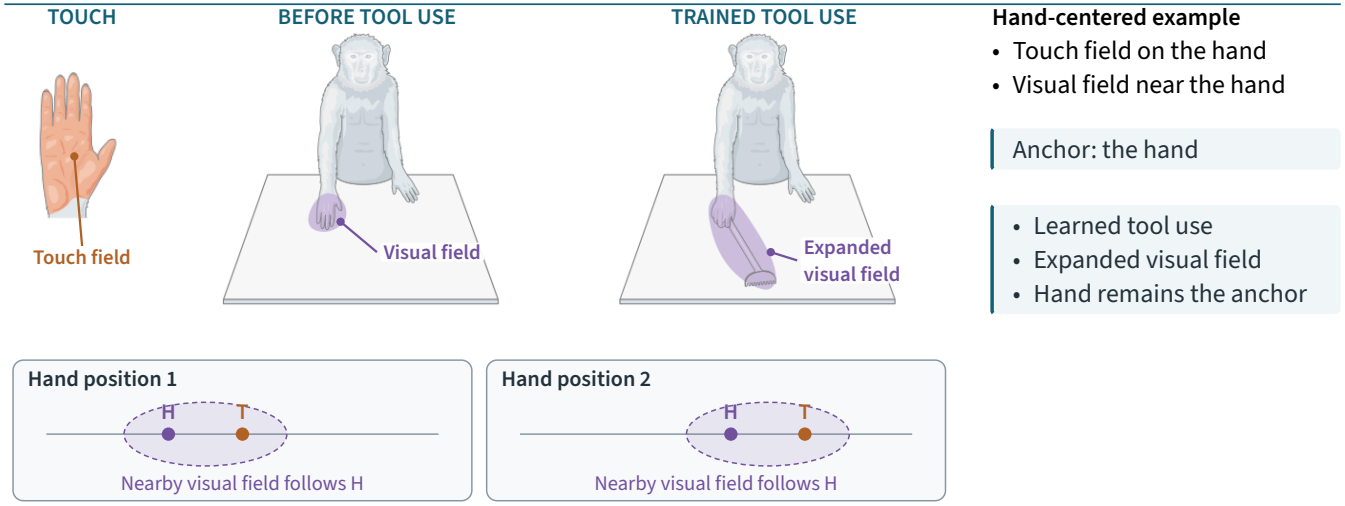
B.09 17 / 50

LECTURE NOTE

This AIP neuron responds to viewing the ring or grasping it in darkness, most strongly when grasping it under vision, but not to another object. Position alone cannot explain the difference: parietal grasp-related activity also relates object shape to the hand configuration needed to grasp it.

YOUR NOTES

Parietal cortex · visual space around the hand



H: hand · T: visual probe · explanatory field, not measured boundaries

Kandel 6e, Fig. 34–6A, p. 828 · receptive-field summary; companion geometry illustrative

Hand-centered example

- Touch field on the hand
- Visual field near the hand

Anchor: the hand

- Learned tool use
- Expanded visual field
- Hand remains the anchor

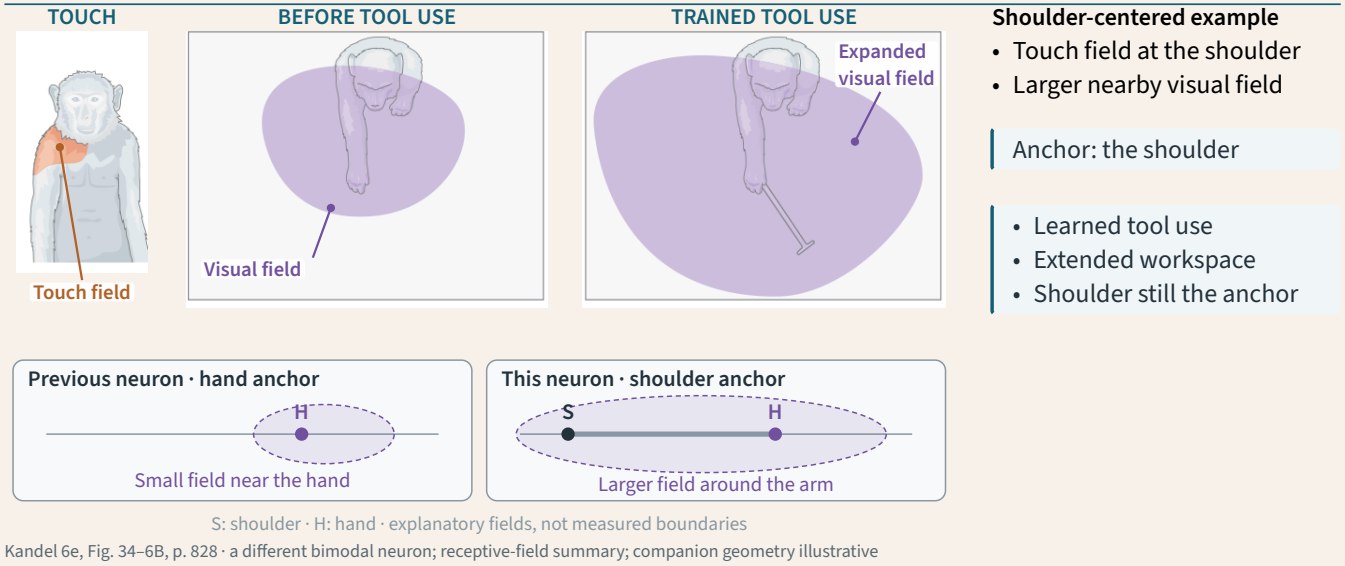
B.10 18 / 50

LECTURE NOTE

Beyond guiding reach and grasp, the hand can anchor sensory space: touch and nearby vision relate to the same hand, so the visual field moves with the arm. The source then shows expansion during use of a previously learned tool; this is a receptive-field summary, not a firing trace or evidence that every held object is incorporated.

YOUR NOTES

Parietal cortex · a shoulder-centered workspace



S: shoulder · H: hand · explanatory fields, not measured boundaries

Kandel 6e, Fig. 34–6B, p. 828 · a different bimodal neuron; receptive-field summary; companion geometry illustrative

Shoulder-centered example

- Touch field at the shoulder
- Larger nearby visual field

Anchor: the shoulder

- Learned tool use
- Extended workspace
- Shoulder still the anchor

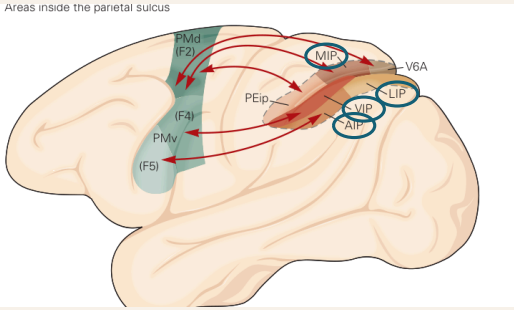
B.11 19 / 50

LECTURE NOTE

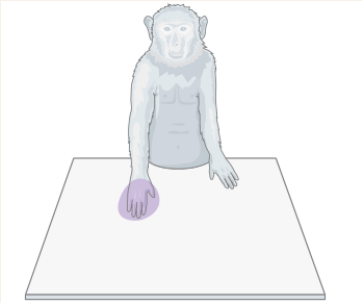
A different illustrated neuron has a shoulder-centered tactile and visual field, covering a larger region than the preceding hand-centered example. Its visual field also expands during learned tool use, extending the idea of body-related space to the workspace available for the whole arm without implying that a hand-centered neuron changed its body anchor.

YOUR NOTES

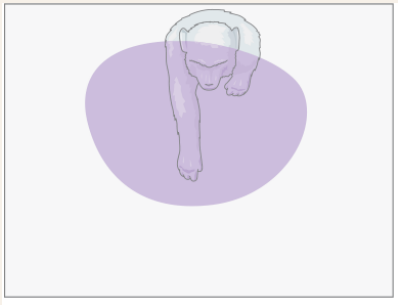
Section B summary: PPC relates body and environment



LIP gaze · VIP head · MIP reach · AIP grasp



Hand-anchored space



Shoulder-anchored space

TAKE HOME PPC relates sensory information to the body for action.

Kandel 6e, Figs. 34–2B (lower), p. 820; 34–6A/B, p. 828 · area experiments in Fig. 34–5

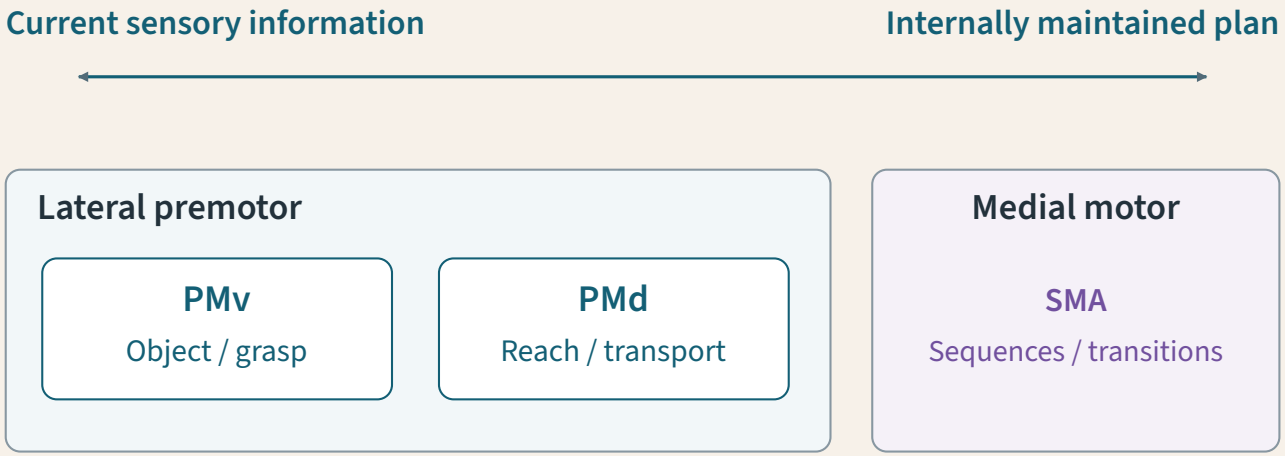
B.12 20 / 50

LECTURE NOTE

Section B showed parietal cortex relating sensory information to the body: stimuli relative to eye or head, reach direction relative to gaze, object shape for grasping, and visual space anchored to the hand or shoulder. These relationships prepare action rather than merely describe sensation, so premotor cortex next takes up how the action itself is organized.

YOUR NOTES

C · What specifies the next action?



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

- Actions combine current sensory guidance and maintained plans.
- These are relative emphases, not separate systems.

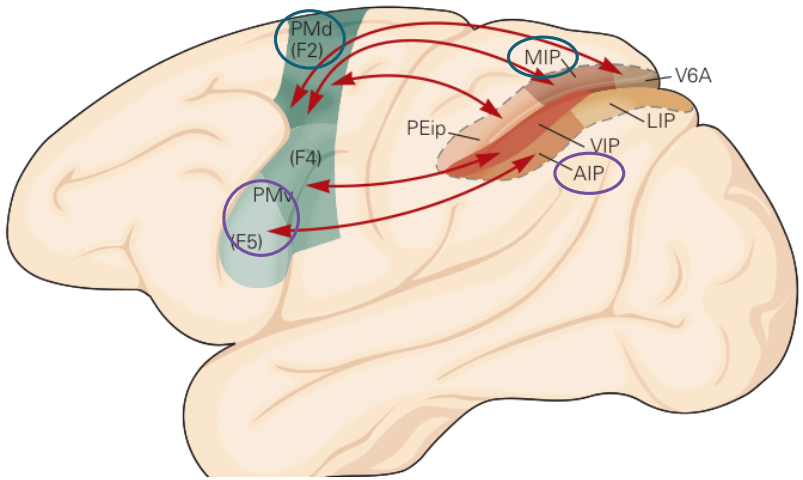
LECTURE NOTE

Premotor contributions can first be organized by the balance of current sensory guidance and internally maintained plans. Reach, grip and remembered sequence provide concrete action themes; these relative emphases overlap and do not divide cortex into externally driven and internally driven compartments.

YOUR NOTES

Locate the connected action circuits

Areas inside the parietal sulcus



- **Reach branch:** MIP ↔ PMd
- **Grasp branch:** AIP ↔ PMv
- Routes run both ways

Kandel 6e, Fig. 34–2B (lower view), p. 820 · parietofrontal pathways; rings added

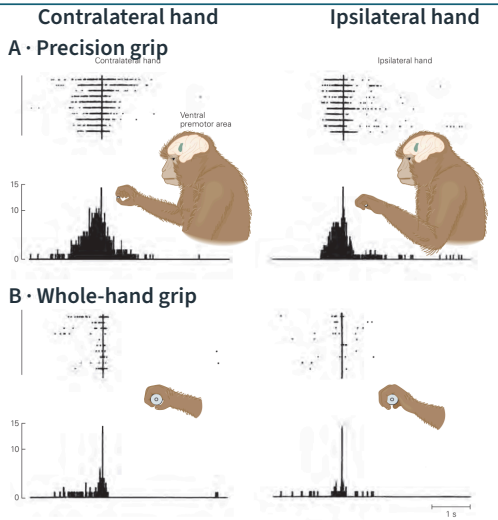
C.02 22 / 50

LECTURE NOTE

The anatomical map reconnects the premotor organizer to PPC. Locate the parietal and frontal reach/grasp branches, including sulcal parietal areas, while preserving reciprocal and cross-circuit connections; region names orient the physiology rather than substitute for it.

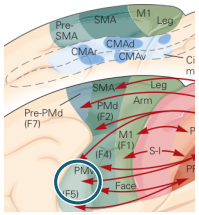
YOUR NOTES

PMv · grip form with either hand



Kandel 6e, Fig. 34–12A/B, p. 837; 34–2B, p. 820 · precision versus whole-hand grip

Ventral premotor area



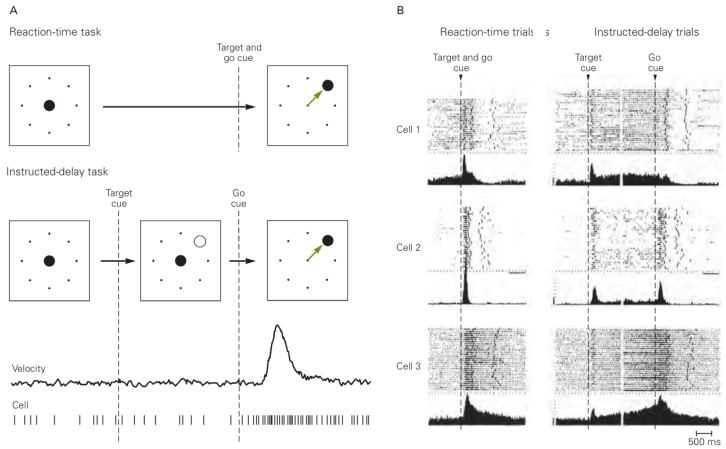
- This cell: precision, both hands
- Other cells prefer other grips
- PMv cells are grip selective

LECTURE NOTE

Compare precision and whole-hand grips, then the two hands within each grip. This PMv neuron prefers precision grip with either hand; other PMv neurons prefer other grip types, precision grip being only the most often represented. The point is grip selectivity that generalizes across hands, developing the grasp branch, not a universal preference or abstract grip code.

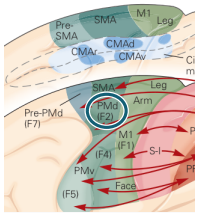
YOUR NOTES

PMd · preparation before movement



Kandel 6e, Fig. 34–4A/B, p. 823; 34–2B, p. 820 · three premotor neurons, recorded in PMd (Crammond & Kalaska, 2000)

Dorsal premotor cortex



- Reaction time: target = go cue
- Instructed delay: target, then go

- Fires after the cue

- Delay: tuned activity before go
- Planning and movement mix
- Cue → maintained reach plan

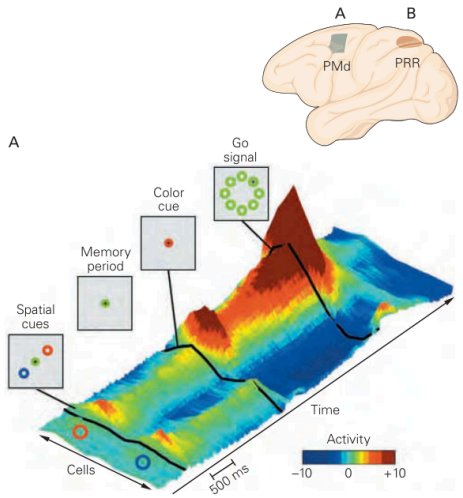
C.04 24 / 50

LECTURE NOTE

The instructed delay separates knowing the target from permission to move. These PMd neurons show directionally tuned delay activity, and single cells mix preparation- and movement-related responses, so an external cue can establish a maintained reach plan; Kandel notes that such delay activity also occurs in other movement-related areas.

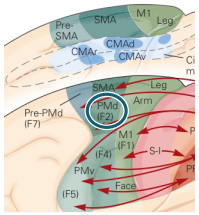
YOUR NOTES

PMd · potential reaches before a choice



Kandel 6e, Fig. 34–8A, p. 832; 34–2B, p. 820 · two target cues, color choice cue and go cue

Dorsal premotor cortex



- Two potential targets: red and blue
- A color cue chooses one
- Go signal: reach

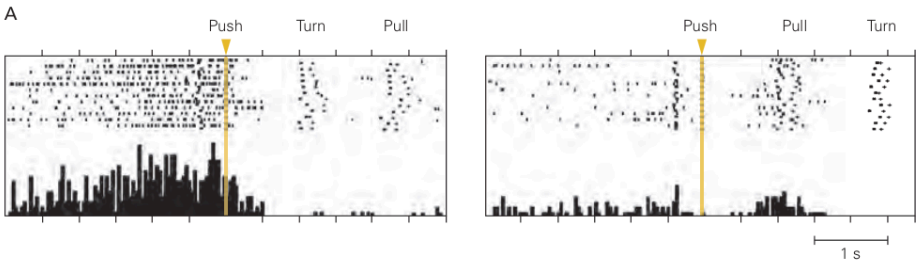
- Delay: both reaches present
- After color cue: chosen reach

LECTURE NOTE

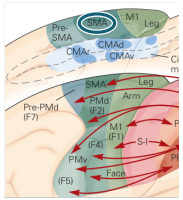
Establish the sequence of target cues, delay, choice and permission to move before reading the population surface. The changing population pattern supports representation of potential actions and selection, but a population average does not show complete alternative plans on every individual trial.

YOUR NOTES

SMA · preparation depends on the remembered sequence



Supplementary motor area



- Memorized sequences
- Both begin with a push

- Push–turn–pull: active before push
- Push–pull–turn: little activity
- Order, not the first act

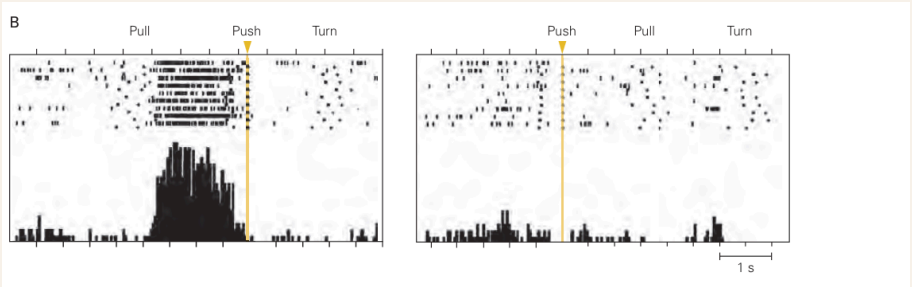
Kandel 6e, Fig. 34–7A, p. 830; 34–2B, p. 820 · sequence comparison; Kandel's category is the supplementary motor complex

LECTURE NOTE

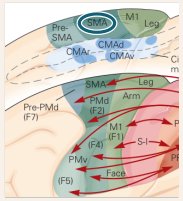
Both compared sequences begin with a push, yet preparation differs with the sequence that follows. The result concerns remembered organization beyond the identity of the immediate movement; Kandel's category is the supplementary motor complex, simplified here to SMA.

YOUR NOTES

SMA · activity at a transition



Supplementary motor area



- Left: pull, then push
- Right: other orders

- Active between pull and push
- Not for push first or pull then turn
- Order organizes the next act

Kandel 6e, Fig. 34–7B, p. 830; 34–2B, p. 820 · sequence transition; Kandel's category is the supplementary motor complex

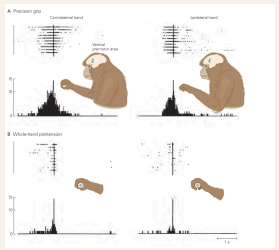
C.07 27 / 50

LECTURE NOTE

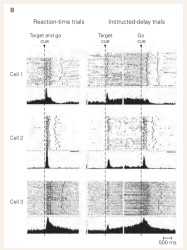
Compare a push after a pull with a push in another sequence position. Activity associated with the transition shows medial motor contributions concerning the organization of acts in context, the maintained-plan side of the organizer, rather than a separate command for each isolated movement.

YOUR NOTES

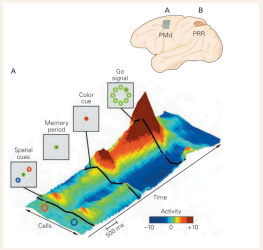
Section C summary: premotor cortex organizes the action



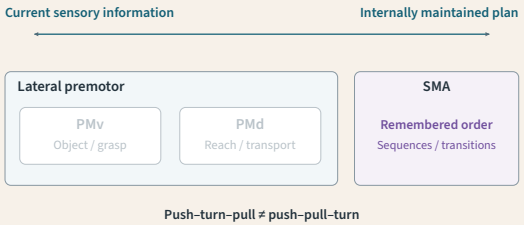
PMv · grip form



PMd · preparation



PMd · potential reaches



SMA · remembered order

TAKE HOME Premotor activity organizes actions before and during movement.

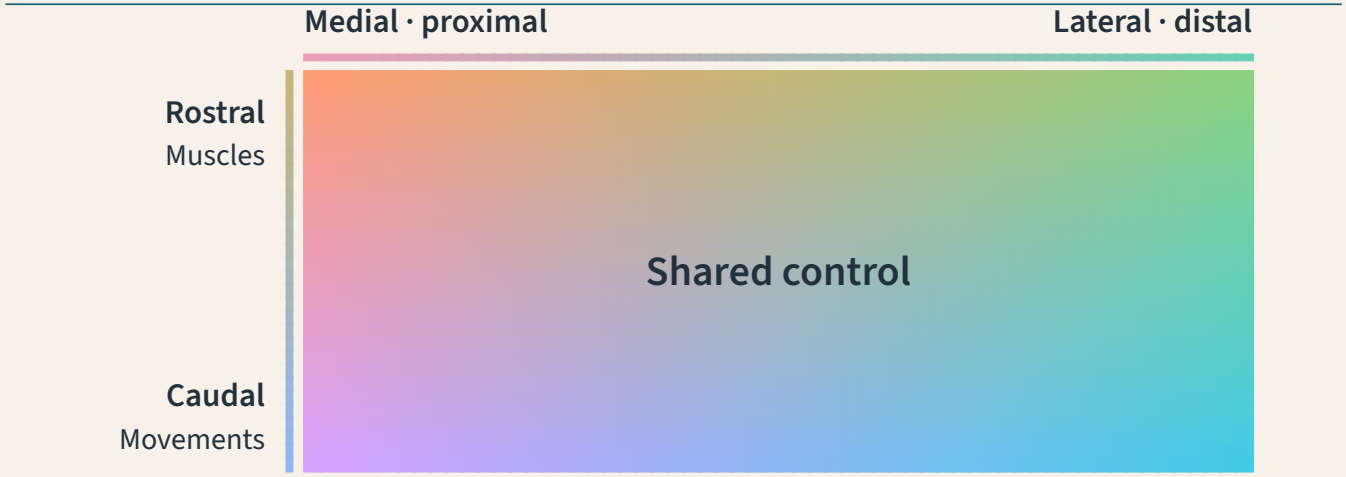
Kandel 6e, Figs. 34–12, 34–4B, 34–8A · sequence schematic after Fig. 34–7

LECTURE NOTE

Section C showed premotor activity organizing an action before and around movement: grip form across hands in PMv, preparation and potential reaches in PMd, and remembered sequences in SMA. These are overlapping contributions within one organizer, balancing sensory guidance against maintained plans; the next section turns to how M1 helps implement the organized action.

YOUR NOTES

D · M1: overlapping control



Kandel 6e, Ch. 34; Kolasinski et al. (2020) · conceptual sketch, not a measured cortical atlas

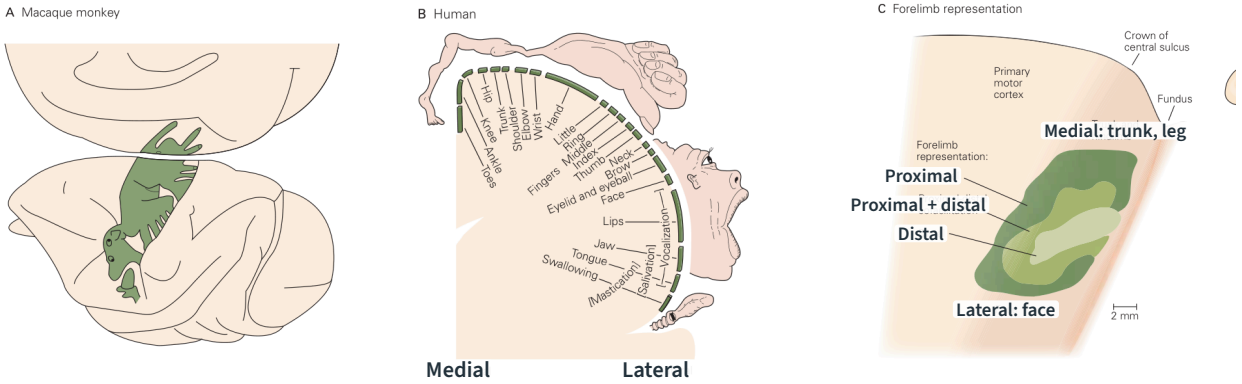
Two overlapping dimensions: which effector, and muscle versus movement emphasis.

LECTURE NOTE

The sketch separates effector control from a functional emphasis that varies along the rostrocaudal dimension in the cited human hand study. It is a conceptual organizer: the textbook anatomy and load experiment do not themselves establish this two-dimensional cortical field.

YOUR NOTES

M1 map · broad order with extensive overlap



- Monkey and human: leg medial, face lateral
- Hand and mouth areas are large

- Distal arm: central core
- Proximal arm: surrounding ring
- Wide overlap between them

Kandel 6e, Fig. 34–16A–C, p. 842 · stimulation-based maps; A macaque, B human, C macaque arm (Park et al., 2001)

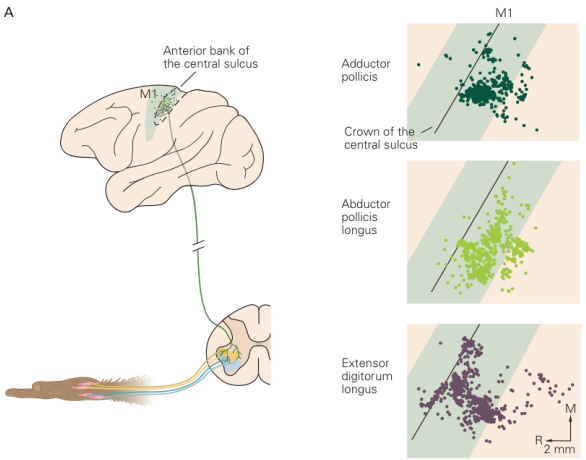
D.02 30 / 50

LECTURE NOTE

Broad body-part order coexists with overlapping forelimb control. Penfield's cartoon shows the relative size of each representation, not separate controllers, and the monkey arm map has a distal core, a proximal surround and a wide zone where both overlap, so the conceptual axis is not a literal medial-to-lateral map of arm muscles.

YOUR NOTES

CM cells · overlapping cortical distributions



Kandel 6e, Fig. 34–17A, p. 843 · CM-cell distributions

Corticomotoneuronal cells

- Direct projection to spinal motor neurons
- Cells for three hand muscles mapped

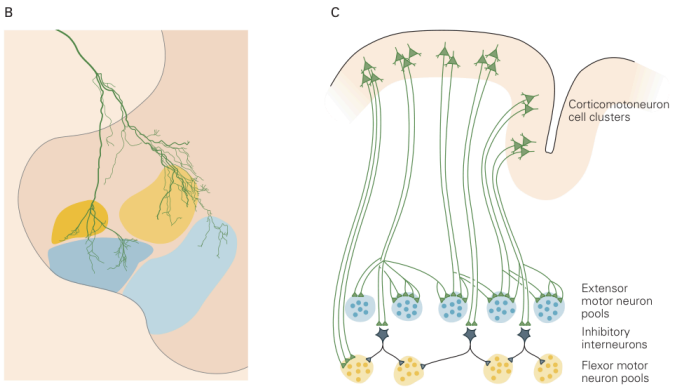
- Caudal M1, anterior bank
- Widely distributed
- Overlapping across muscles

LECTURE NOTE

CM cells provide direct access to spinal motor neurons and are concentrated in caudal M1 in this anatomy. Cells associated with different muscles overlap extensively; this projection pattern does not by itself determine what movement variable their activity represents.

YOUR NOTES

CM output · combinations of muscles



Corticomotoneuronal output

- B: one CM axon in the ventral horn
- C: direct and interneuron routes

- One axon reaches several motor pools
- Outputs combine muscles

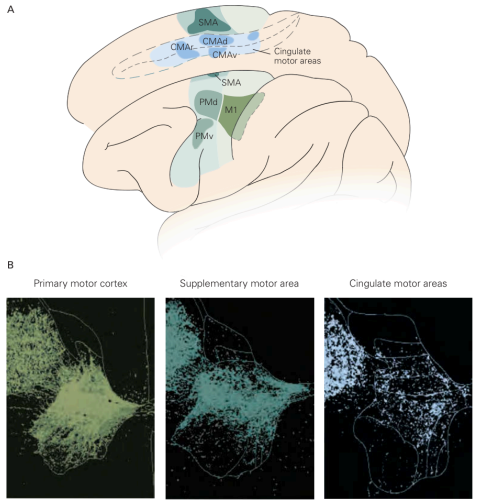
Kandel 6e, Fig. 34–17B/C, p. 843 · divergent terminals and spinal routes

LECTURE NOTE

A CM axon can influence several motor pools and interneuronal circuits. Direct and indirect routes help explain coordinated muscle combinations, while spinal motor neurons integrate other descending and local influences before producing the final output.

YOUR NOTES

Descending control · M1 among several sources



Kandel 6e, Fig. 34–3A/B, p. 822 · cortical origins and spinal terminations

Corticospinal origins

- A: arm and hand origins
- B: spinal terminations

- Many frontal areas project
- M1 is one of several sources

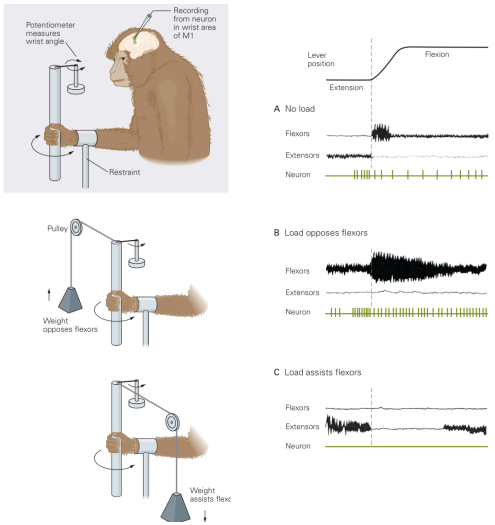
D.05 33 / 50

LECTURE NOTE

Several cortical regions have descending projections with different terminations. M1’s distinctive direct access coexists with other routes, so the muscle combinations reached by CM cells sit within a shared descending controller rather than an isolated final cortical stage.

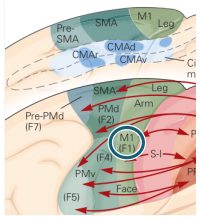
YOUR NOTES

M1 · same displacement, different muscle output



Kandel 6e, Fig. 34–18A–C, p. 846; 34–2B, p. 820 · no load, opposing load and assisting load

Primary motor cortex, wrist area



- Same wrist flexion each time
- A: no load · B: load opposes flexors · C: load assists

- Flexor EMG follows the load
- So does the neuron
- Activity tracks force, not displacement

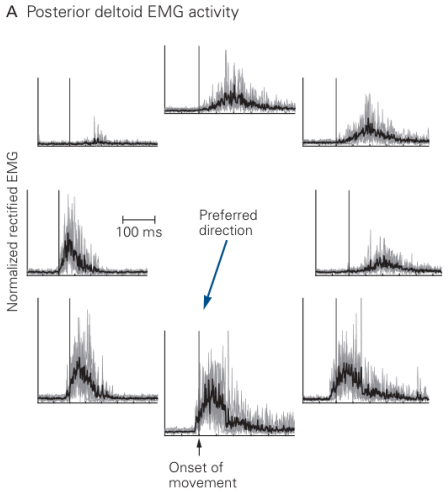
D.06 34 / 50

LECTURE NOTE

Read the mechanical setup before comparing displacement, muscle activity and neural firing. The illustrated neuron changes with the force and muscle demands while displacement remains similar, dissociating visible movement from the output required to produce it without establishing a universal M1 force code.

YOUR NOTES

Muscle · broad tuning to reach direction



Kandel 6e, Fig. 34–20A, p. 848 · posterior deltoid EMG

Posterior deltoid

- Reaches in eight directions
- EMG aligned on movement onset

- Peak near 250°
- Also active nearby

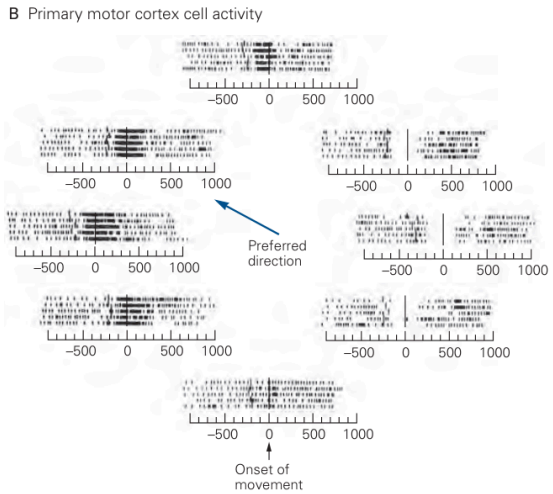
D.07 35 / 50

LECTURE NOTE

A muscle is most active for a preferred reach direction but contributes across neighboring directions. This supplies an essential interpretation check: a directional tuning curve alone cannot establish that a neural response exclusively encodes the hand's trajectory.

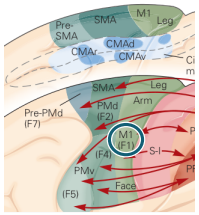
YOUR NOTES

M1 neuron · broad tuning to reach direction



Kandel 6e, Fig. 34–20B, p. 848; 34–2B, p. 820 · example M1 neuron

Primary motor cortex



- The same eight reach directions

- Maximal near 135–180°
- Lower across other directions
- Lowest opposite the preferred

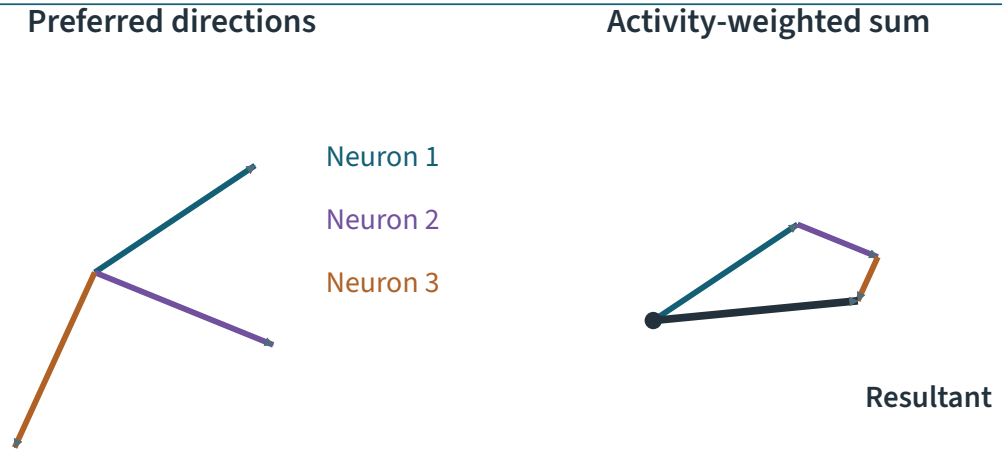
D.08 36 / 50

LECTURE NOTE

Compare the same neuron across the eight reach directions. Its preferred direction marks maximal activity rather than an exclusive movement command; many broadly tuned neurons contribute to each reach, motivating a population-level description.

YOUR NOTES

Weight preferred directions, then add the vectors



Kandel 6e, pp. 849–850 · illustrative three-neuron weighted sum, not recorded values

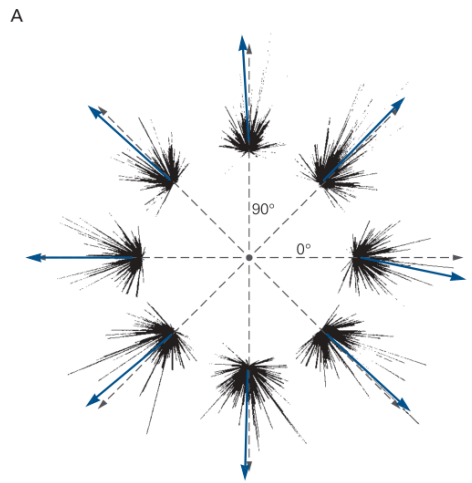
- Each vector: a preferred direction, scaled by firing
- Their sum: the population vector

LECTURE NOTE

Assign each neuron a vector along its preferred direction and scale its length by the neuron’s activity. Adding these contributions produces a resultant; the illustration explains the analysis rather than claiming that neurons literally implement this vector addition.

YOUR NOTES

Population vector · same cells, different resultants



Kandel 6e, Fig. 34–22A, p. 850 · thin cell vectors, blue population vectors and dashed reach directions

M1 population

- Thin lines: single-cell vectors
- Blue: population vector
- Dashed: reach direction

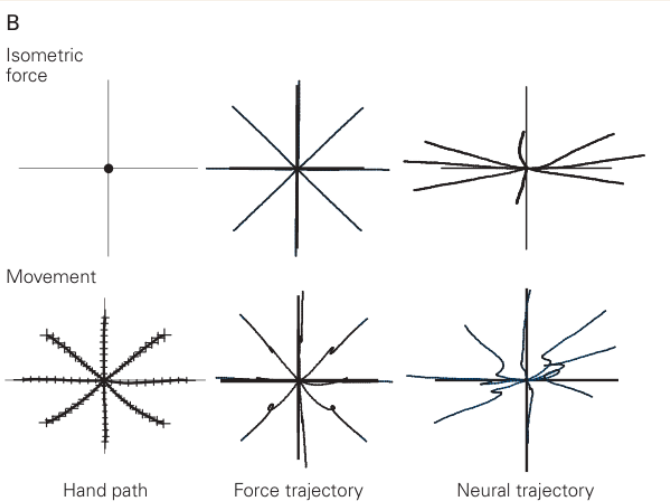
- Vector ≈ reach direction
- Same cells, different weights

LECTURE NOTE

Read one cluster before comparing all eight: activity-weighted cell vectors sum to the blue population vector. Different activity weights in the same distributed population distinguish reach directions, demonstrating useful decoding without establishing one invariant neural representation.

YOUR NOTES

Population vector · not a pure hand code



Kandel 6e, Fig. 34–22B, p. 850 · hand path, force trajectory and neural trajectory

M1 population

- Top: isometric force, no movement
- Bottom: moving a heavy handle
- Compare hand, force and neural columns

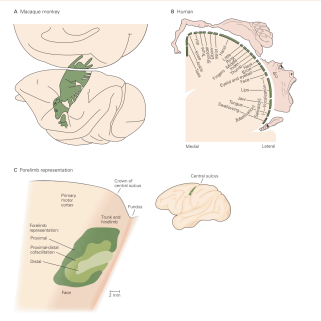
- Varies with no hand motion
- They reverse while braking
- Reflects task mechanics

LECTURE NOTE

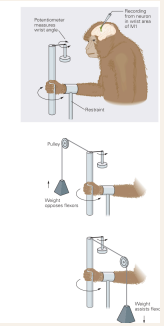
Compare the isometric and moving-handle rows by columns. Neural vectors vary without hand motion and can reverse during braking while the hand continues toward its target, showing that task mechanics and force demands matter for interpreting the decoded signal.

YOUR NOTES

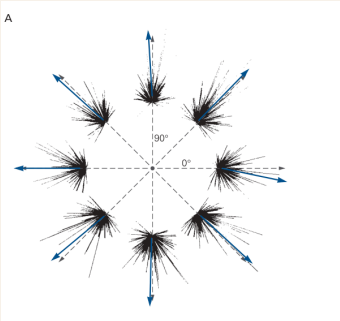
Section D summary: M1 coordinates output for the body and task



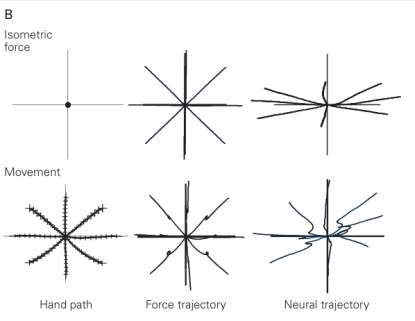
Overlapping map



Force demands



Population vector



Not a pure hand code

TAKE HOME M1 control is overlapping, distributed and shaped by force demands.

Kandel 6e, Figs. 34–16, p. 842; 34–18, p. 846; 34–22A/B, p. 850

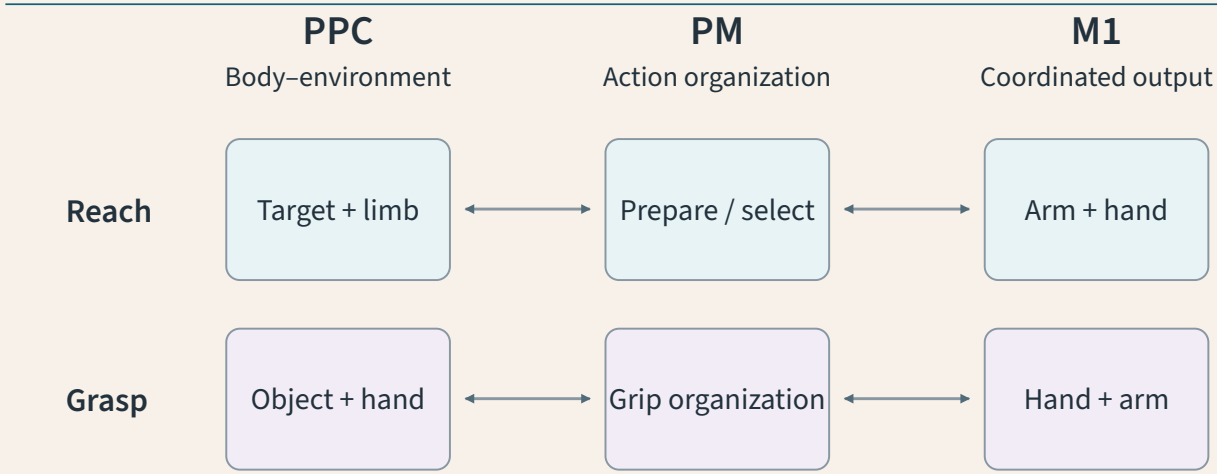
D.12 40 / 50

LECTURE NOTE

Section D showed M1 as overlapping, distributed control: cortical outputs reach muscle combinations, activity follows force demands as well as displacement, and a population vector decodes reach direction without being a pure hand-direction code. M1 works within shared descending control, which sets up the final question of how this controller changes with experience.

YOUR NOTES

E · Learning changes a connected controller



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

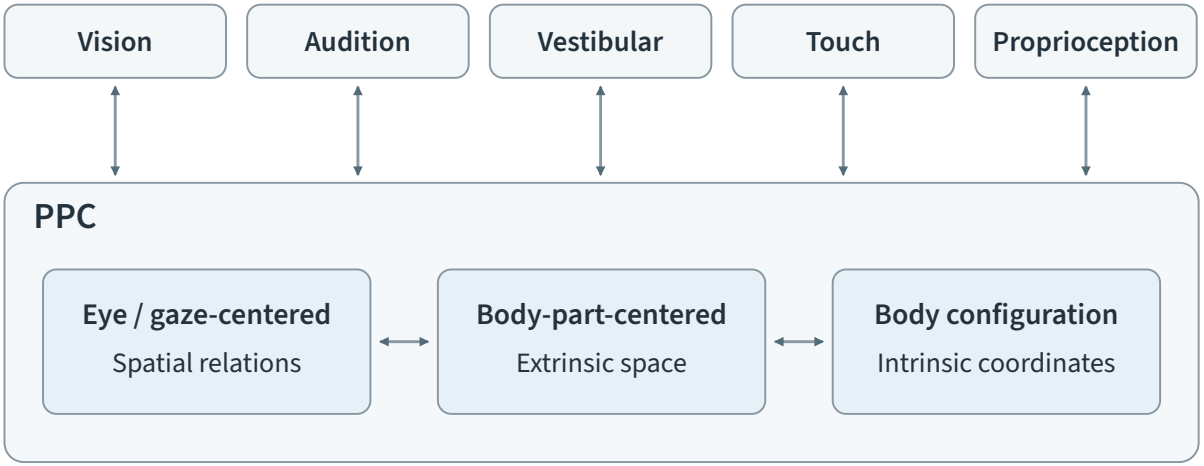
Learning can change the connected controller developed in sections A–D.

LECTURE NOTE

The overall network now serves as a recap rather than a new theory. Recall how body–environment relationships support organized action and coordinated output, then ask which relationships or output capabilities could change with experience.

YOUR NOTES

Recall PPC's integrated body–environment relationships



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

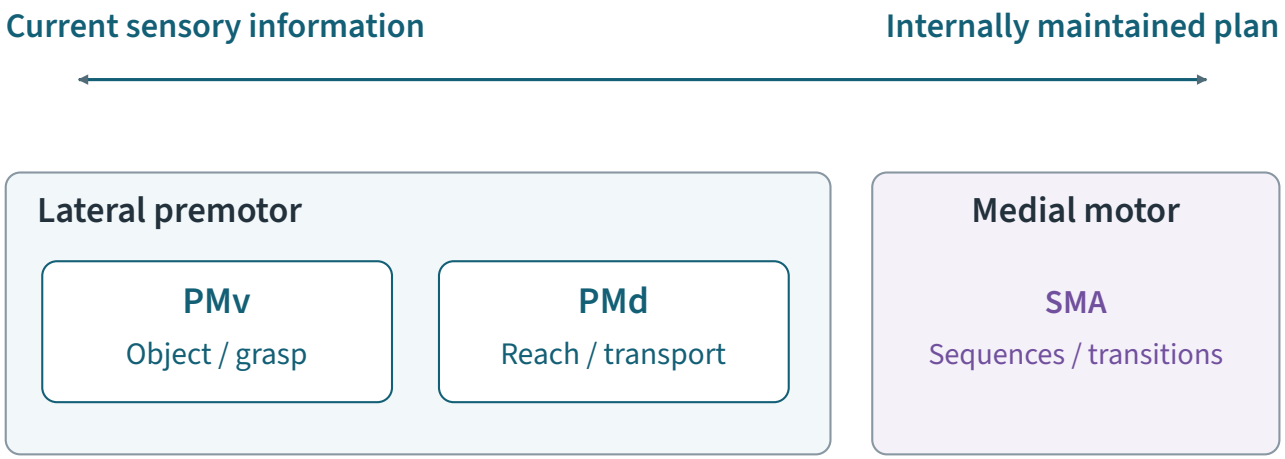
E.02 42 / 50

LECTURE NOTE

PPC relates several sensory and body-state descriptions for action. This recap brings its contribution back into the connected controller before the learning examples, without claiming that the following M1 experiments measure changes in every PPC representation.

YOUR NOTES

Recall premotor action organization



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

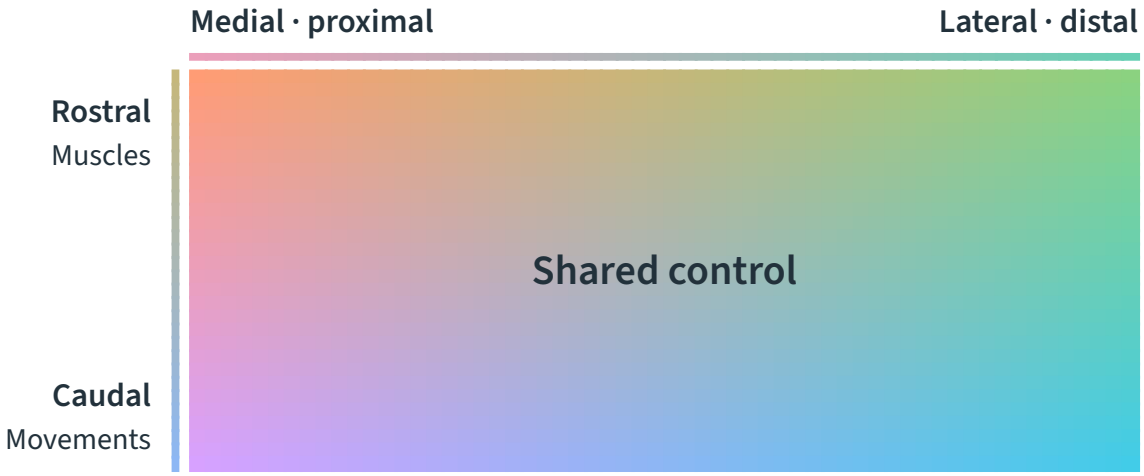
E.03 43 / 50

LECTURE NOTE

Premotor contributions connect sensory guidance, maintained information and organized action forms. Recalling these themes makes learning a question about changing coordinated control rather than merely enlarging a fixed anatomical area.

YOUR NOTES

Recall M1’s overlapping control



Kandel 6e, Ch. 34; Kolasinski et al. (2020) · conceptual sketch; retain the anatomy and coding distinctions

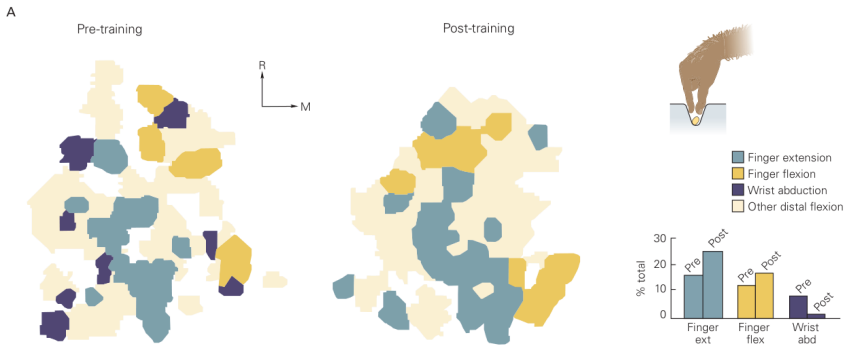
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LECTURE NOTE

M1 contributes through distributed populations and overlapping access to motor output. The conceptual sketch recalls that organization, while the next experiment asks a more directly testable question: does practice alter what cortical stimulation can evoke?

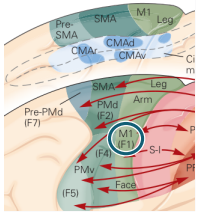
YOUR NOTES

M1 map · practice changes the evoked movements



Kandel 6e, Fig. 34–24A, p. 854; 34–2B, p. 820 · pre-training and post-training maps

Primary motor cortex



- Training: retrieve pellets from a small well
- Map evoked movements, before training

> - Finger areas expand
> - Wrist abduction shrinks

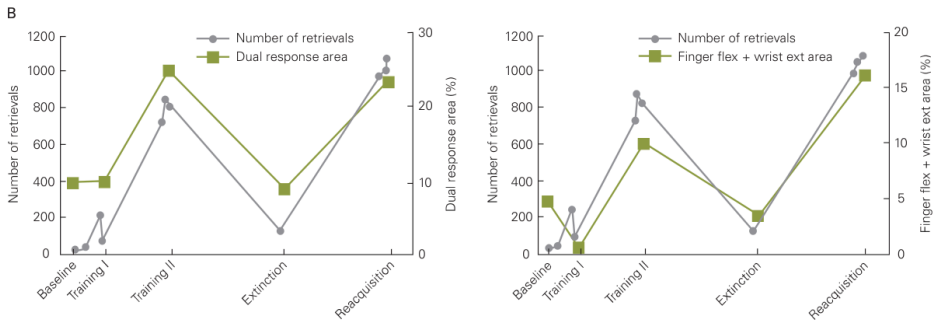
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LECTURE NOTE

Establish the pellet-retrieval skill, then compare the evoked finger and wrist movements before and after training. The colors identify movement categories rather than firing strength; the changed output map shows experience-dependent organization without demonstrating new neurons or a complete learning mechanism.

YOUR NOTES

M1 map · changes track skill acquisition and loss



Primary motor cortex

- Retrievals and mapped area across training, extinction and retraining
- A different monkey from the maps

Kandel 6e, Fig. 34–24B, p. 854 · performance and mapped areas in a different monkey

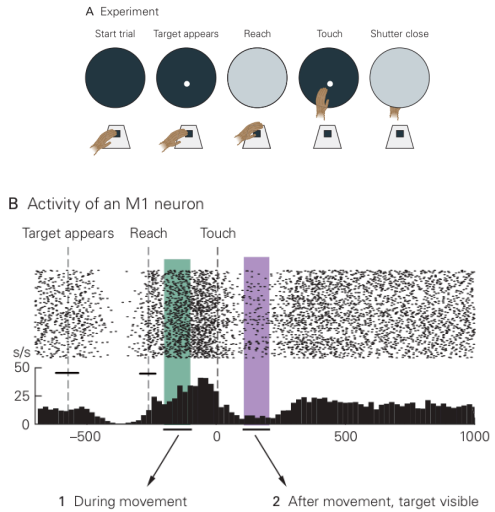
- Map area tracks performance
- Different measures, same course

LECTURE NOTE

Read both vertical axes and compare training, loss of practice and reacquisition. Retrieval performance and the mapped response areas vary together but measure different quantities; this animal also differs from the one in the preceding before-and-after maps.

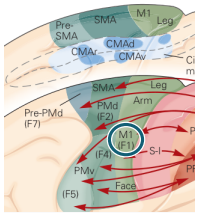
YOUR NOTES

M1 · movement versus feedback about its result



Kandel 6e, Fig. 34–25A/B, p. 856; 34–2B, p. 820 · prism shift, shutter and brief endpoint feedback; example M1 neuron

Primary motor cortex



- Prisms shift the seen target
- Shutter: no view during the reach
- Brief view of the endpoint after touch

- Green: during movement
- Purple: after feedback
- Next: relate each to the error

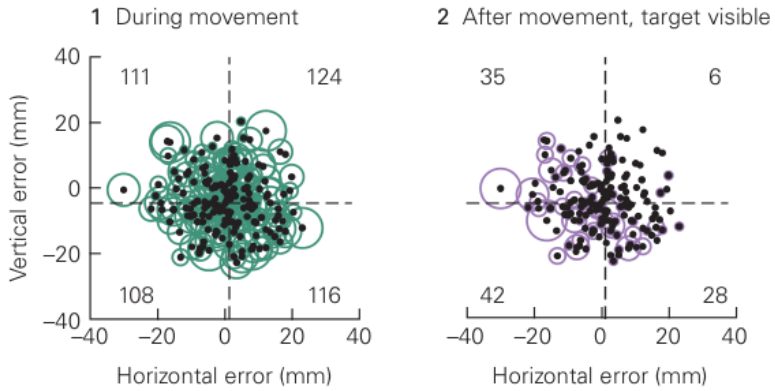
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LECTURE NOTE

The shutter prevents visual feedback during the reach, and a brief view then reveals the endpoint relative to the target. This separates activity while producing the movement (green epoch) from activity after information about its error becomes available (purple epoch), which the next slide relates to the size and direction of the error.

YOUR NOTES

M1 neuron · error direction after feedback



Kandel 6e, Fig. 34–25B1/B2, p. 856 · endpoint errors and firing-rate circles

Primary motor cortex

- Dots: endpoint errors from the target
- Circle size: firing in the epoch

- During movement: no relation
- After feedback: error direction

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LECTURE NOTE

Black dots show endpoint errors and circle diameter represents firing in the selected epoch. The directional relationship emerges after feedback in this example; it does not prove that this neuron computes the error alone or that all adaptation occurs in M1.

YOUR NOTES

Testing a contribution to learning requires an intervention

OBSERVE
Error-related activity after feedback

INTERVENE
Simulate the activity with cortical stimulation

TEST
Does the next reach change?

> The text reports adaptive changes even without an actual reaching error.

Kandel 6e, p. 855 · described M1/PMd intervention; separate from Fig. 34–25's recorded responses.

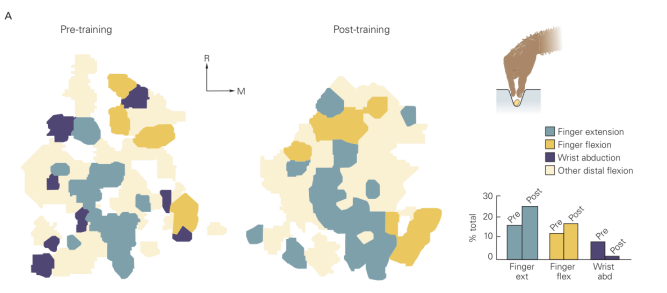
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LECTURE NOTE

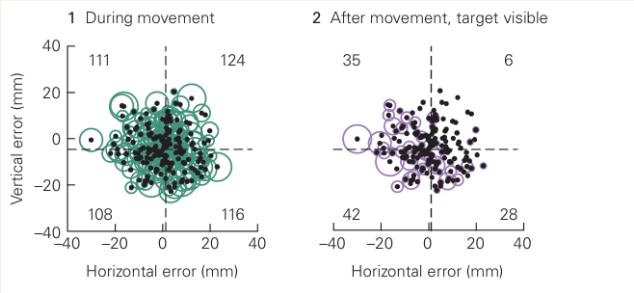
Kandel describes a separate stimulation test that simulated error-related responses in M1 and PMd and biased subsequent reaches despite no actual error. This intervention strengthens the link to learning, while remaining distinct from the recording correlations displayed in the preceding figure.

YOUR NOTES

Section E summary: the controller changes with experience



Practice reshapes the map



Error direction after feedback

TAKE HOME Practice and error feedback change the cortical controller.

Kandel 6e, Figs. 34–24A, p. 854; 34–25B, p. 856

LECTURE NOTE

Section E showed that the controller is not fixed: practice reshapes the output evoked from the M1 map, and activity after feedback can carry an error's direction, with a stimulation test strengthening the link to learning. The lecture has moved from relating body and environment, through organizing an action, to producing and refining it.

YOUR NOTES